

Harvard EDU S043 Final Project - Exploratory Analysis

Andrew Bogdan

Lin Jin

Livia Cheng

2025-11-22

Workspace

This is setting up to use a virtual environment. It should print something like “The library is already synchronized with the lockfile.”.

```
renv::restore()
```

- The library is already synchronized with the lockfile.

```
-- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
v dplyr      1.1.4      v readr      2.1.5
v forcats    1.0.1      v stringr    1.6.0
v ggplot2    4.0.1      v tibble     3.3.0
v lubridate  1.9.4      v tidyr      1.3.1
v purrr      1.1.0
```

```
-- Conflicts ----- tidyverse_conflicts() --
```

```
x dplyr::filter() masks stats::filter()
```

```
x dplyr::lag()     masks stats::lag()
```

```
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become
```

```
Loading required package: Matrix
```

Attaching package: 'Matrix'

The following objects are masked from 'package:tidyr':

expand, pack, unpack

```
source("util.R")

data <- readRDS("data/clean_data.rds")
nrow(data) # We expect 325883
```

```
[1] 325883
```

Exploratory Data Analysis

To Do 1. Unconditional Random Intercept Model 2. Conditional Random Intercept Model (with all covariates) 3. Use these two models to generate plots - We need to know if the models have to demonstrate clustering somehow. - We can make a heatmap of each of our groups; 27 x 20. - We need to include average number of people per group - Our groups have *wildly* different numbers of people in them - How many people in the dataset? How many with **mismatch**?

There are 51834 datapoints in our sample which represent people without jobs. They are missing `job_satis`, `job_type`, and so on— these people are not useful to us, so we will drop them into space.

```
data <- data %>% drop_na(job_satis)
```

Here's an idea: what if our groups are pairs of major and job type?

```
data <- data %>%
  group_by(major, job_type) %>%
  mutate(pair_id = cur_group_id()) %>%
  ungroup()
```

Plots

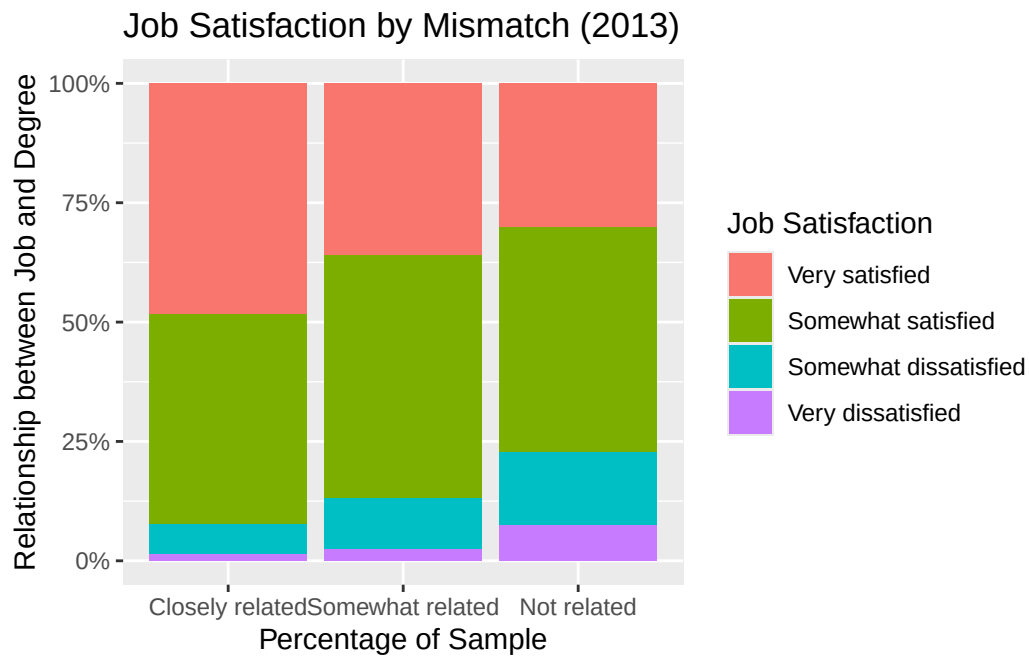
Let's do a basic plot to see what the odds

```
# Here we use the latest measurement for each person in our sample.
# I think the dataset is so large that this decision doesn't really matter.
data %>%
  filter(survey_year == 2013) %>%
  ggplot(aes(x = as_factor(mismatch), fill = as_factor(job_satis))) +
  geom_bar(position = "fill") +
  labs(
    x = "Percentage of Sample",
```

```

y = "Relationship between Job and Degree",
fill = "Job Satisfaction",
title = "Job Satisfaction by Mismatch (2013)"
) +
scale_y_continuous(labels = scales::percent_format())

```



```

data %>%
  filter(survey_year == 2013) %>%
  ggplot(aes(y = as_factor(reason_1_for_mismatch), fill = as_factor(job_satis))) +
  geom_bar(position = "fill") +
  labs(
    x = "Reason",
    y = "Percentage of Sample",
    fill = "Job Satisfaction",
    title = "Job Satisfaction by Reason"
  ) +
  scale_x_continuous(labels = scales::percent_format())

```



ICC

Okay, let's take a look at the ICC.

```
model <- cache(
  expression = lmer(job_satis ~ (1 | person_id) + (1 | pair_id), data = data),
  cache_file = "cache/pairs_unconditional_model.rds",
  use_cache = TRUE
)
```

```
[1] "Loading from cache..."
```

```
vars <- get_variance(model)
vars / sum(vars)
```

```
person_id pair_id Residual
0.37009653 0.01307839 0.61682508
```

```
model <- cache(
  expression = lmer(job_satis ~ (1 | pair_id), data = data %>% filter(survey_year == 2013)),
  cache_file = "cache/unconditional_pairs_2013.rds",
  use_cache = TRUE
)
```

```
[1] "Loading from cache..."
```

```
vars <- get_variance(model)
vars / sum(vars)
```

```
pair_id    Residual
0.02006298 0.97993702
```

Basic Model

```
# Let's make our factors nicer
lm_data <- data %>%
  mutate(
    job_satis = 4 - job_satis,
    # I like the idea of using a monotone coding or something, but with
    # only three elements on the scale, I don't think it matters
    mismatch = mismatch %>% as_factor() %>% fct_recode(
      ":C" = "Closely related", ":S" = "Somewhat related", ":N" = "Not related"
    ),
    reason = reason_1_for_mismatch %>% as_factor() %>%
      fct_relevel("Logical Skip") %>%
      fct_recode(
        ":Pay" = "Pay, promotion opportunities",
        ":Conds" = "Working conditions",
        ":Loc" = "Job location",
        ":Change" = "Change in career or professional interests",
        ":Fam" = "Family-related reasons",
        ":Job" = "Job in highest degree field not available",
        ":Other" = "Other reason for not working",
        ":NA" = "Logical Skip"
      ),
    highest_degree = highest_degree %>% as_factor() %>% fct_recode(
      ":B" = "Bachelor's", ":M" = "Master's", ":Dr" = "Doctorate",
      ":Pr" = "Professional"
    ),
    # salary =
    # Our sample is mostly men (about 60%) so I'm setting them as the default
    gender = gender %>% as_factor() %>% fct_relevel("Male") %>%
      fct_recode(":M" = "Male", ":F" = "Female"),
    # I'm centering this around 36 to 40 hours per week, as that's full time.
```

```

    weekly_hours - 3,
    # Our data is 65% white people
    race = race %>% as_factor() %>% fct_relevel("White") %>%
      fct_recode(":W" = "White", ":A" = "Asian", ":M" = "Under-represented minorities")
  )

model <- lmer(
  job_satis ~ 1 +
    mismatch +
    # reason +
    highest_degree +
    # salary # There are too many artifacts in this data to use it right now
    weekly_hours +
    gender +
    age +
    race +
    # (1 | person_id) + # Random intercept for each person
    (1 | pair_id), # Random intercept for each major + job type pair
  data = lm_data %>% filter(survey_year == 2013)
)

get_variance(model) / sum(get_variance(model))

```

```

      pair_id    Residual
0.009725188 0.990274812

```

```
get_variance(model)
```

```

      pair_id    Residual
0.004854742 0.494337897

```

```
broom.mixed::tidy(model)
```

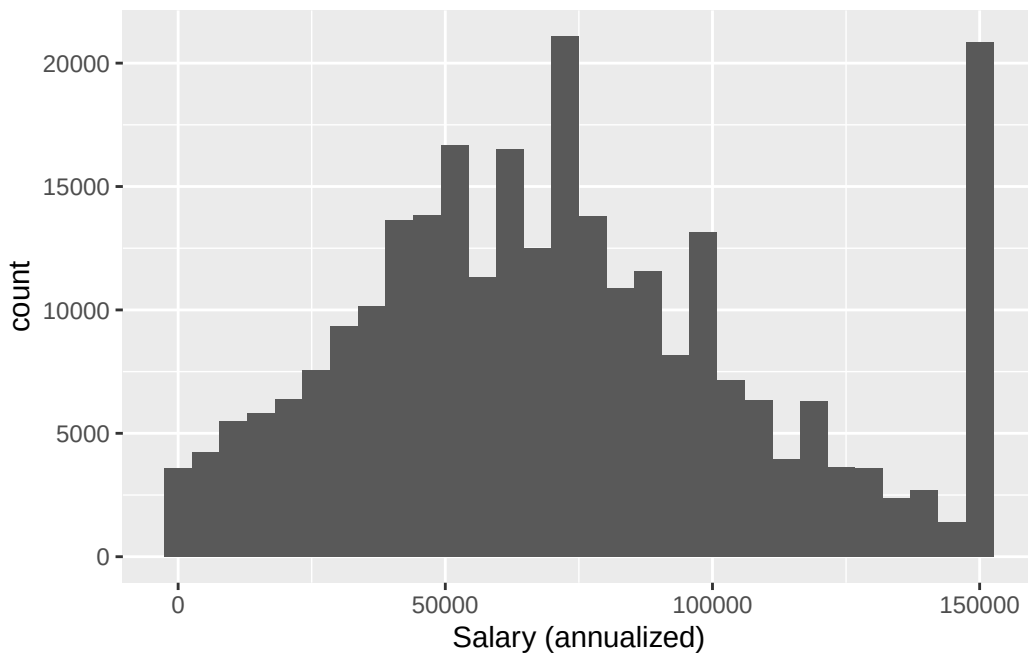
```

# A tibble: 13 x 6
  effect    group    term          estimate std.error statistic
  <chr>    <chr>    <chr>          <dbl>    <dbl>    <dbl>
1 fixed   <NA>    (Intercept)    2.23     0.0167    133.
2 fixed   <NA>    mismatch:S    -0.194    0.00672   -28.9
3 fixed   <NA>    mismatch:N    -0.389    0.00899   -43.3

```

4	fixed	<NA>	highest_degree:M	-0.00137	0.00606	-0.226
5	fixed	<NA>	highest_degree:Dr	-0.00346	0.0156	-0.222
6	fixed	<NA>	highest_degree:Pr	0.0462	0.0145	3.19
7	fixed	<NA>	weekly_hours	0.00603	0.00315	1.91
8	fixed	<NA>	gender:F	-0.00811	0.00594	-1.37
9	fixed	<NA>	age	0.00465	0.000219	21.3
10	fixed	<NA>	race:A	-0.0983	0.00759	-13.0
11	fixed	<NA>	race:M	-0.0634	0.00665	-9.53
12	ran_pars	pair_id	sd__(Intercept)	0.0697	NA	NA
13	ran_pars	Residual	sd__Observation	0.703	NA	NA

```
data %>%
  ggplot(aes(x = salary)) +
  geom_histogram(bins = 30)
```



Ideas

For ordinal covariates, we could use monotone coding. It seems like it adds a parameter which allows the scales to be spaced differently apart. For example, it might transform a Likert Scale (0, 1, 2) onto (0, 1.2, 3.5) because it seems like 1-2 is a more extreme gap than 0-1.

For linear-ish covariates which saturate (e.g. salary), we could use a tobit approach, which seems to add a parameter which estimates the chance of saturating. I'm not sure how it works.

For the ordinal outcome, we should use ordinal regression.

Citations

```
citation("tidyverse")
```

To cite package 'tidyverse' in publications use:

Wickham H, Averick M, Bryan J, Chang W, McGowan LD, François R, Grolemund G, Hayes A, Henry L, Hester J, Kuhn M, Pedersen TL, Miller E, Bache SM, Müller K, Ooms J, Robinson D, Seidel DP, Spinu V, Takahashi K, Vaughan D, Wilke C, Woo K, Yutani H (2019). "Welcome to the tidyverse." *_Journal of Open Source Software_*, *4*(43), 1686. doi:10.21105/joss.01686 <<https://doi.org/10.21105/joss.01686>>.

A BibTeX entry for LaTeX users is

```
@Article{,
  title = {Welcome to the {tidyverse}},
  author = {Hadley Wickham and Mara Averick and Jennifer Bryan and Winston Chang and Lucy L},
  year = {2019},
  journal = {Journal of Open Source Software},
  volume = {4},
  number = {43},
  pages = {1686},
  doi = {10.21105/joss.01686},
}
```

```
citation("lme4")
```

To cite lme4 in publications use:

Douglas Bates, Martin Maechler, Ben Bolker, Steve Walker (2015). Fitting Linear Mixed-Effects Models Using lme4. *Journal of Statistical Software*, 67(1), 1-48. doi:10.18637/jss.v067.i01.

A BibTeX entry for LaTeX users is

```
@Article{,
  title = {Fitting Linear Mixed-Effects Models Using {lme4}},
  author = {Douglas Bates and Martin M{"a}chler and Ben Bolker and Steve Walker},
  journal = {Journal of Statistical Software},
  year = {2015},
  volume = {67},
  number = {1},
  pages = {1--48},
  doi = {10.18637/jss.v067.i01},
}
```

- Wickham H, Averick M, Bryan J, Chang W, McGowan LD, François R, Grolemund G, Hayes A, Henry L, Hester J, Kuhn M, Pedersen TL, Miller E, Bache SM, Müller K, Ooms J, Robinson D, Seidel DP, Spinu V, Takahashi K, Vaughan D, Wilke C, Woo K, Yutani H (2019). “Welcome to the tidyverse.” *Journal of Open Source Software*, 4(43), 1686. doi:10.21105/joss.01686 <https://doi.org/10.21105/joss.01686>.
- Douglas Bates, Martin Maechler, Ben Bolker, Steve Walker (2015). Fitting Linear Mixed-Effects Models Using lme4. *Journal of Statistical Software*, 67(1), 1-48. doi:10.18637/jss.v067.i01.